HOST J1: connect to Raspberry Pi 4 3V3, GND, GPIO2/SDA1, GPIO3/SCL1.

picOrick ultrasound acquisition is a separate USB module; no high-voltage path is on this hub.

Three I2C switches: 0x70, 0x71, 0x72; each active branch has one TMP102 node at 0x48.

Connector footprints, cable routing, total 3V3 load and skin-contact implementation require review.

U1
TCA9548APWR 0x70

U2
TCA9548APWR 0x71

U3
TCA9548APWR 0x72

J1
HOST_3V3_I2C

#FLG01
PWR_FLAG

#FLG02
PWR_FLAG

C1
100n

R1
10k

C2
100n

R14
10k

C3
100n

R27
10k

J2
SENSOR_01

J3
SENSOR_02

J4
SENSOR_03

J5
SENSOR_04

J6
SENSOR_05

J7
SENSOR_06

J8
SENSOR_07

J9
SENSOR_08

J10
SENSOR_09

J11
SENSOR_10

J12
SENSOR_11

J13
SENSOR_12

J14
SENSOR_13

J15
SENSOR_14

J16
SENSOR_15

J17
SENSOR_16

J18
SENSOR_17

J19
SENSOR_18

R2 4k7
R3 4k7
R4 4k7
R5 4k7
R6 4k7
R7 4k7
R8 4k7
R9 4k7
R10 4k7
R11 4k7
R12 4k7
R13 4k7
R15 4k7
R16 4k7
R17 4k7
R18 4k7
R19 4k7
R20 4k7
R21 4k7
R22 4k7
R23 4k7
R24 4k7
R25 4k7
R26 4k7
R28 4k7
R29 4k7
R30 4k7
R31 4k7
R32 4k7
R33 4k7
R34 4k7
R35 4k7
R36 4k7
R37 4k7
R38 4k7
R39 4k7

Research prototype: electrical design requires review

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Title: Mammary Protect — 18-position temperature I2C hub

Size: A2	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/1